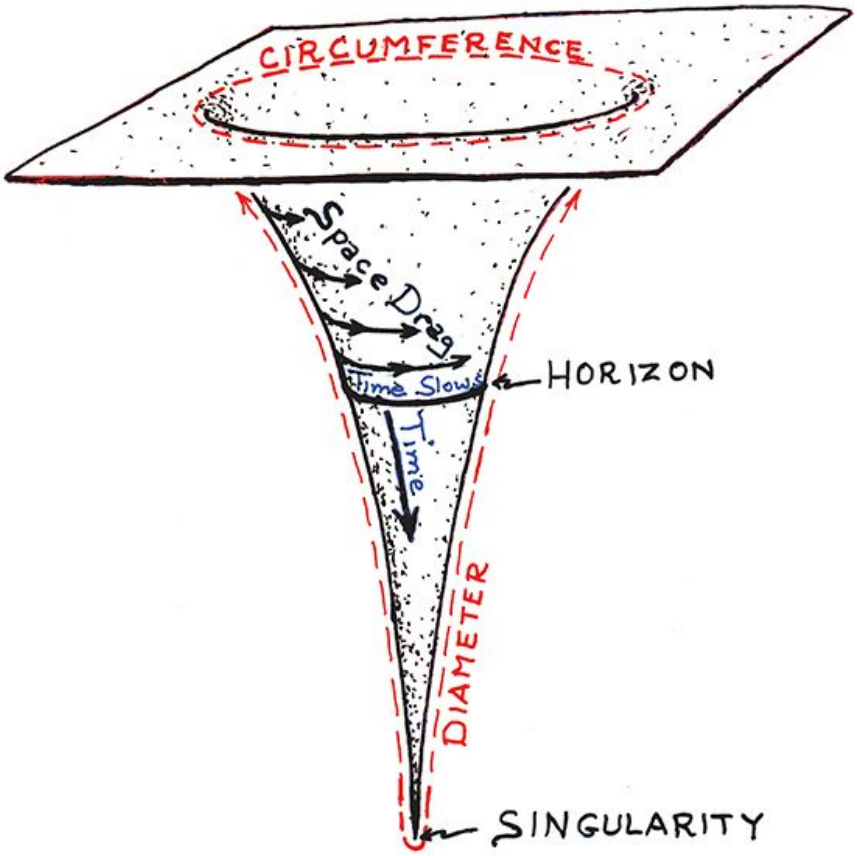


Fig. 5.4. Space around a spinnning black hole is dragged into whirling motion. *[My own hand sketch.]*

Precise Depiction of the Warped Space and Time Around a Black Hole

These three aspects of spacetime warping—the warp of space, the slowing and distortion of time, and the whirl of space—are all described by mathematical formulas. These formulas have been deduced from Einstein's relativistic laws, and their precise predictions are depicted quantitatively in Figure 5.5 (by contrast with Figures 5.1–5.4, which were only qualitative).

The warped shape of the surface in Figure 5.5 is precisely what we would see from the bulk, when looking at the hole's equatorial plane. The colors depict the slowing of time as measured by someone who